

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CS-4004-CBGS

B.E. IV Semester

Examination, June 2020

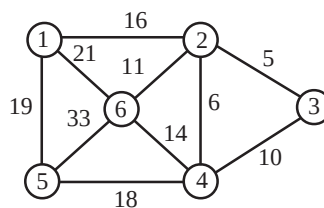
Choice Based Grading System (CBGS) Analysis and Design of Algorithm

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume any data suitably.

1. a) Explain Strassen's algorithm for matrix multiplication with an example. 7
b) Illustrate Heap Sort in the array 7
A = [5, 8, 3, 9, 2, 10, 1, 40]
2. a) Explain job sequencing problem with example. 7
b) Obtain a set of optimal Huffman codes for the seven messages ($M_1, \dots, M_7$) with relative frequencies ($q_1, \dots, q_7$) = (4, 5, 7, 8, 10, 22, 15). Draw the decode tree for this set of codes. 7
3. a) Consider the Knapsack instance $n = 3$, $(w_1, w_2, w_3) = (2, 3, 4)$ and $(p_1, p_2, p_3) = (1, 2, 5)$ and $m = 5$. Find the optimal solution. 7
b) Find the shortest path for vertex ① to vertex ③ in the following graph using Dijkstra's greedy algorithm. 7



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4. a) What is multistage graph problem? Discuss its solution based on dynamic programming approach. Also give a suitable algorithm and find its computing time? 7
b) Explain Hamiltonian cycle and write an algorithm to find all Hamiltonian cycle in graph. 7
5. a) Explain Dynamic Programming concept with example. 7
b) What is principle of optimality. Give suitable example. 7
6. a) What is branch and bound techniques? 7
b) Create a B-tree for the following list of elements. 7
(86, 50, 40, 3, 94, 10, 70, 90, 110, 113, 116). Give minimization factor $t = 3$, min degree = 2, max degree = 5.
7. a) Explain the difference between Big - Oh (O), Omega (ω) and theta (θ) notation. 7
b) Explain 2-3 tree with the help of suitable example. 7
8. Write short notes on: 14
 - a) Parallel Algorithm
 - b) NP completeness
 - c) Reliability Design
 - d) Merge Sort
